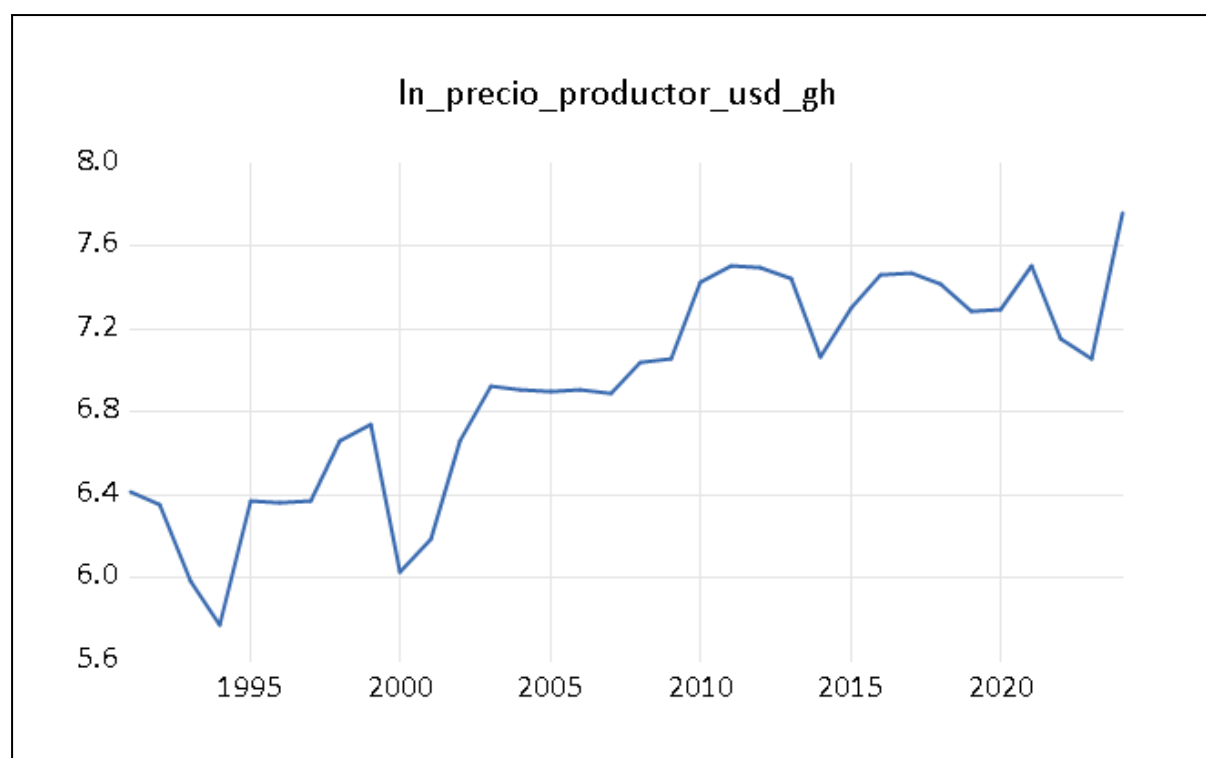




Date: 09/27/26 Time: 19:15

Sample: 1991 2024

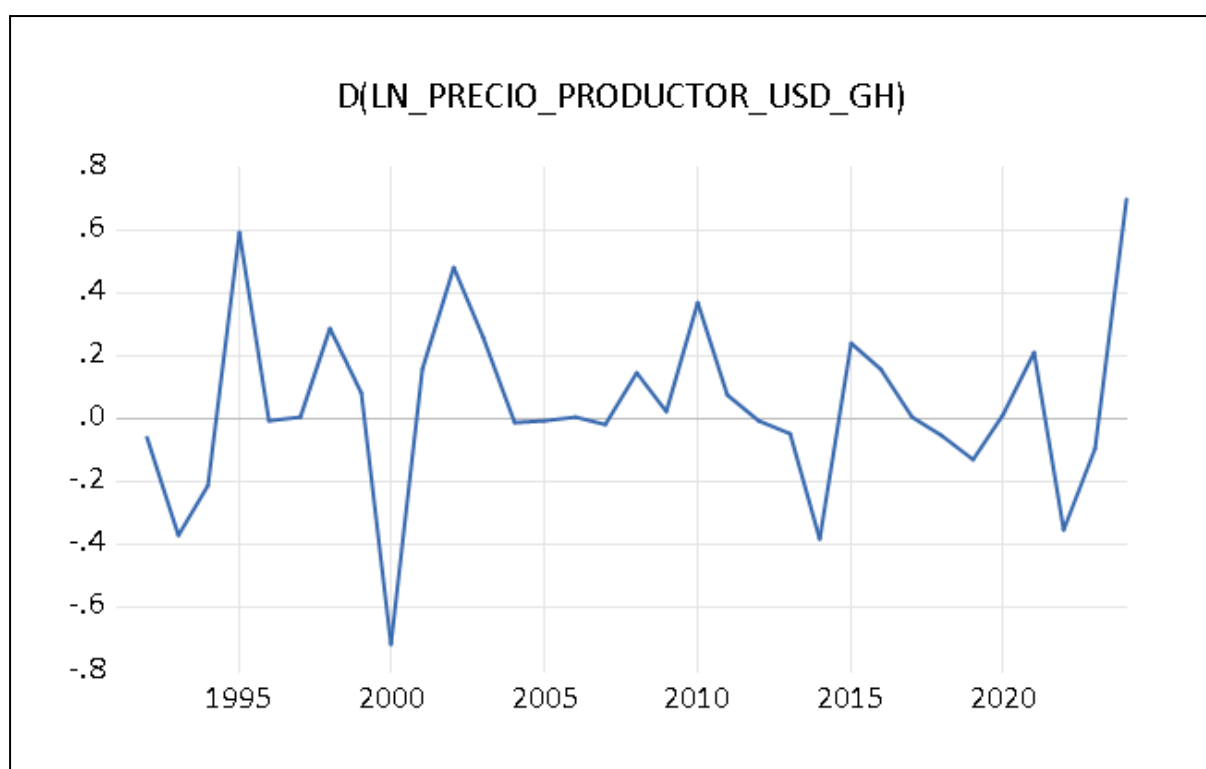
Included observations: 34

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
1		0.802	0.802	23.828	0.000
2		0.675	0.091	41.255	0.000
3		0.635	0.197	57.163	0.000
4		0.536	-0.112	68.868	0.000
5		0.484	0.095	78.772	0.000
6		0.474	0.079	88.591	0.000
7		0.426	-0.015	96.812	0.000
8		0.330	-0.165	101.95	0.000
9		0.202	-0.237	103.95	0.000
10		0.048	-0.275	104.07	0.000
11		-0.008	0.081	104.07	0.000
12		-0.055	-0.028	104.24	0.000
13		-0.074	0.130	104.56	0.000
14		-0.127	-0.178	105.55	0.000
15		-0.238	-0.193	109.20	0.000
16		-0.310	-0.025	115.72	0.000

Utilizo el test de Dickey Fuller Ampliado para verificar la presencia de raíz unitaria antes de estimar modelos.

Null Hypothesis: LN_PRECIO_PRODUCTOR_USD_GH has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 0 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-3.441470	0.0631
Test critical values:	1% level		-4.262735	
	5% level		-3.552973	
	10% level		-3.209642	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH)				
Method: Least Squares				
Date: 09/27/26 Time: 19:19				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LN_PRECIO_PRODUCTOR_USD_GH(-1)	-0.555030	0.161277	-3.441470	0.0017
C	3.416802	0.994224	3.436654	0.0017
@TREND("1991")	0.026369	0.008381	3.146306	0.0037
R-squared	0.286207	Mean dependent var		0.040716
Adjusted R-squared	0.238621	S.D. dependent var		0.278510
S.E. of regression	0.243020	Akaike info criterion		0.095162
Sum squared resid	1.771762	Schwarz criterion		0.231208
Log likelihood	1.429826	Hannan-Quinn criter.		0.140937
F-statistic	6.014495	Durbin-Watson stat		1.676522
Prob(F-statistic)	0.006362			

Los coeficientes de intercepto y tendencia dan significativos por lo que miro el tau critico al 5%, como el test no rechaza la presencia de raiz unitaria al 5% (-3.44>-3.55) entonces diferencio la serie una vez.



Nuevamente utilizo ADF test para buscar raíces unitarias.

Null Hypothesis: D(LN_PRECIO_PRODUCTOR_USD_GH) has a unit root				
Exogenous: Constant, Linear Trend				
Lag Length: 1 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-6.371415	0.0001
Test critical values:	1% level		-4.284580	
	5% level		-3.562882	
	10% level		-3.215267	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH,2)				
Method: Least Squares				
Date: 10/01/26 Time: 18:03				
Sample (adjusted): 1994 2024				
Included observations: 31 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_PRECIO_PRODUCTOR_USD_GH(-1	-1.612771	0.253126	-6.371415	0.0000
D(LN_PRECIO_PRODUCTOR_USD_GH(-1	0.531894	0.177333	2.999409	0.0058
C	0.106291	0.102764	1.034326	0.3102
@TREND("1991")	-0.001926	0.005084	-0.378760	0.7078
R-squared	0.627734	Mean dependent var		0.034424
Adjusted R-squared	0.586372	S.D. dependent var		0.391636
S.E. of regression	0.251877	Akaike info criterion		0.200162
Sum squared resid	1.712933	Schwarz criterion		0.385192
Log likelihood	0.897495	Hannan-Quinn criter.		0.260477
F-statistic	15.17629	Durbin-Watson stat		2.029280
Prob(F-statistic)	0.000006			

Como el coeficiente de tendencia determinista no es significativo pruebo el ADF test con intercepto.

Null Hypothesis: D(LN_PRECIO_PRODUCTOR_USD_GH) has a unit root				
Exogenous: Constant				
Lag Length: 1 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-6.464377	0.0000
Test critical values:	1% level		-3.661661	
	5% level		-2.960411	
	10% level		-2.619160	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH,2)				
Method: Least Squares				
Date: 10/01/26 Time: 18:04				
Sample (adjusted): 1994 2024				
Included observations: 31 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_PRECIO_PRODUCTOR_USD_GH(-1	-1.603453	0.248044	-6.464377	0.0000
D(LN_PRECIO_PRODUCTOR_USD_GH(-1	0.528673	0.174399	3.031406	0.0052
C	0.071414	0.044914	1.590014	0.1231
R-squared	0.625756	Mean dependent var		0.034424
Adjusted R-squared	0.599025	S.D. dependent var		0.391636
S.E. of regression	0.247994	Akaike info criterion		0.140945
Sum squared resid	1.722035	Schwarz criterion		0.279718
Log likelihood	0.815357	Hannan-Quinn criter.		0.186181
F-statistic	23.40879	Durbin-Watson stat		2.026680
Prob(F-statistic)	0.000001			

Como el coeficiente del intercepto tampoco es significativo utilizo el ADF test sin coeficientes.

Null Hypothesis: D(LN_PRECIO_PRODUCTOR_USD_GH) has a unit root				
Exogenous: None				
Lag Length: 1 (Automatic - based on SIC, maxlag=8)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-6.152366	0.0000
Test critical values:	1% level		-2.641672	
	5% level		-1.952066	
	10% level		-1.610400	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH,2)				
Method: Least Squares				
Date: 10/01/26 Time: 18:04				
Sample (adjusted): 1994 2024				
Included observations: 31 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LN_PRECIO_PRODUCTOR_USD_GH(-1	-1.552768	0.252386	-6.152366	0.0000
D(LN_PRECIO_PRODUCTOR_USD_GH(-1	0.502661	0.178145	2.821632	0.0085
R-squared	0.591966	Mean dependent var		0.034424
Adjusted R-squared	0.577895	S.D. dependent var		0.391636
S.E. of regression	0.254445	Akaike info criterion		0.162873
Sum squared resid	1.877519	Schwarz criterion		0.255388
Log likelihood	-0.524533	Hannan-Quinn criter.		0.193031
Durbin-Watson stat	1.895909			

Al ser el último test simplemente observo el estadístico asociado al tau al 5% de significatividad, como $-6.15 < -1.95$ rechazo la hipótesis nula de raíz unitaria, por lo que debo estimar sobre esta serie diferenciada una vez.

Date: 10/01/26 Time: 18:03						
Sample (adjusted): 1992 2024						
Included observations: 33 after adjustments						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	-0.043	-0.043	0.0663	0.797
		2	-0.433	-0.436	7.0645	0.029
		3	0.016	-0.036	7.0743	0.070
		4	0.004	-0.230	7.0750	0.132
		5	-0.160	-0.242	8.1372	0.149
		6	0.071	-0.092	8.3499	0.214
		7	0.190	0.001	9.9586	0.191
		8	0.115	0.169	10.574	0.227
		9	-0.023	0.135	10.599	0.304
		10	-0.260	-0.142	14.002	0.173
		11	-0.081	-0.039	14.350	0.214
		12	-0.081	-0.341	14.712	0.258
		13	0.097	-0.035	15.254	0.292
		14	0.211	-0.035	17.964	0.208
		15	-0.024	-0.124	18.001	0.263
		16	-0.096	-0.049	18.627	0.288

Tanto en la Función de Autocorrelación como en la Parcial solo en el segundo rezago los estadísticos son significativos.

Podría tratarse de un ARIMA(2,1,2)

Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 10/01/26 Time: 18:28				
Sample (adjusted): 1994 2024				
Included observations: 31 after adjustments				
Failure to improve likelihood (non-zero gradients) after 21 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1992 1993				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.073731	0.046569	1.583242	0.1250
@TREND	-0.001657	0.002357	-0.703274	0.4879
AR(2)	0.082310	0.185977	0.442584	0.6616
MA(2)	-0.929411	0.061191	-15.18859	0.0000
R-squared	0.463271	Mean dependent var		0.057105
Adjusted R-squared	0.403635	S.D. dependent var		0.276678
S.E. of regression	0.213664	Akaike info criterion		-0.128913
Sum squared resid	1.232608	Schwarz criterion		0.056117
Log likelihood	5.998156	Hannan-Quinn criter.		-0.068598
F-statistic	7.768253	Durbin-Watson stat		2.523776
Prob(F-statistic)	0.000677			
Inverted AR Roots	.29	-.29		
Inverted MA Roots	.96	-.96		

Sacando la tendencia:

Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 10/01/26 Time: 18:29				
Sample (adjusted): 1994 2024				
Included observations: 31 after adjustments				
Failure to improve likelihood (non-zero gradients) after 25 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1992 1993				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.041766	0.009561	4.368411	0.0002
AR(2)	0.096273	0.158507	0.607374	0.5485
MA(2)	-0.940509	0.053602	-17.54625	0.0000
R-squared	0.451899	Mean dependent var		0.057105
Adjusted R-squared	0.412748	S.D. dependent var		0.276678
S.E. of regression	0.212025	Akaike info criterion		-0.172461
Sum squared resid	1.258726	Schwarz criterion		-0.033688
Log likelihood	5.673152	Hannan-Quinn criter.		-0.127225
F-statistic	11.54271	Durbin-Watson stat		2.429503
Prob(F-statistic)	0.000221			
Inverted AR Roots	.31	-.31		
Inverted MA Roots	.97	-.97		

Saco el componente AR(2):

Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH)				
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)				
Date: 10/01/26 Time: 18:08				
Sample (adjusted): 1992 2024				
Included observations: 33 after adjustments				
Failure to improve likelihood (non-zero gradients) after 11 iterations				
Coefficient covariance computed using outer product of gradients				
MA Backcast: 1990 1991				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.043465	0.017470	2.487943	0.0184
MA(2)	-0.635557	0.125897	-5.048247	0.0000
R-squared	0.276104	Mean dependent var		0.040716
Adjusted R-squared	0.252752	S.D. dependent var		0.278510
S.E. of regression	0.240754	Akaike info criterion		0.048611
Sum squared resid	1.796839	Schwarz criterion		0.139308
Log likelihood	1.197924	Hannan-Quinn criter.		0.079128
F-statistic	11.82383	Durbin-Watson stat		2.331012
Prob(F-statistic)	0.001689			
Inverted MA Roots	.80	-.80		

Ambos coeficientes son significativos y las raíces están dentro del círculo unitario.

Date: 10/01/26 Time: 18:08
Sample (adjusted): 1992 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.205	-0.205	1.5095	
		2 0.002	-0.042	1.5096	0.219
		3 -0.122	-0.136	2.0844	0.353
		4 0.105	0.054	2.5243	0.471
		5 -0.172	-0.157	3.7513	0.441
		6 0.140	0.071	4.5873	0.468
		7 0.126	0.186	5.2904	0.507
		8 0.023	0.059	5.3153	0.622
		9 0.014	0.104	5.3248	0.722
		10 -0.275	-0.288	9.1135	0.427
		11 -0.078	-0.206	9.4359	0.491
		12 -0.101	-0.194	9.9953	0.531
		13 0.024	-0.192	10.028	0.613
		14 0.154	0.160	11.477	0.571
		15 -0.090	-0.153	11.995	0.607
		16 -0.025	0.000	12.037	0.676

El residuo es ruido blanco, todas las barras están dentro de las bandas de significatividad y la probabilidad de los test Liung-Box son todas mayores a 0.05.

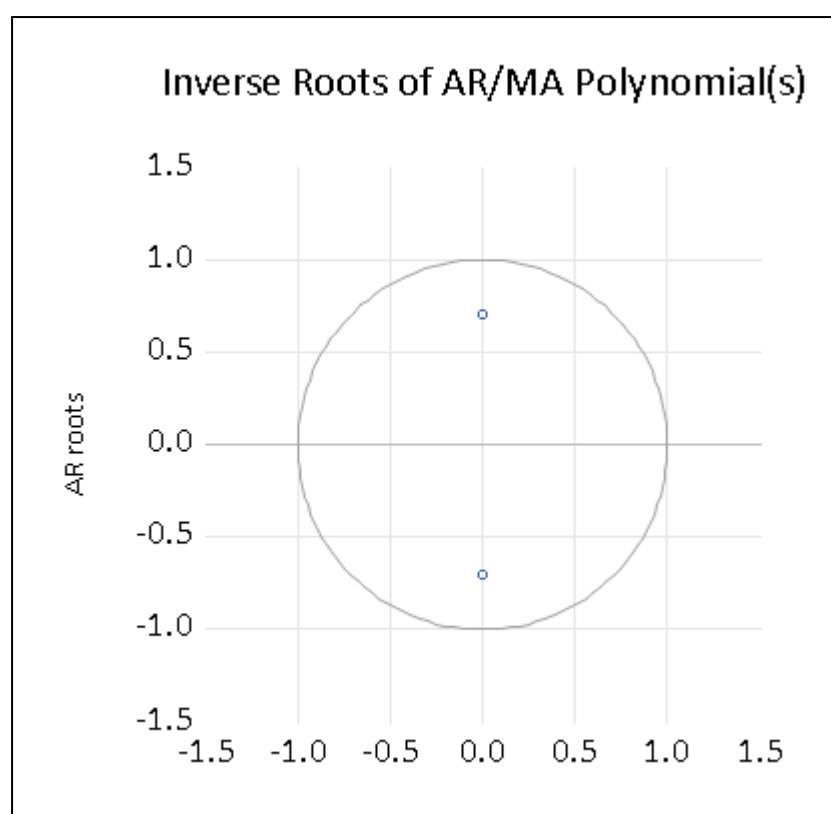
Sin embargo, pruebo con la estimación de un ARIMA(2,1,0) ya que a veces ambos coeficientes juntos no dan significativos pero sí por separado.

Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH)
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)

Date: 10/01/26 Time: 18:06
Sample (adjusted): 1994 2024
Included observations: 31 after adjustments
Convergence achieved after 3 iterations
Coefficient covariance computed using outer product of gradients

Variable	Coefficient	Std. Error	t-Statistic	Prob.
AR(2)	-0.502413	0.175388	-2.864573	0.0076
R-squared	0.180214	Mean dependent var		0.057105
Adjusted R-squared	0.180214	S.D. dependent var		0.276678
S.E. of regression	0.250510	Akaike info criterion		0.101089
Sum squared resid	1.882655	Schwarz criterion		0.147346
Log likelihood	-0.566876	Hannan-Quinn criter.		0.116168
Durbin-Watson stat	1.978485			
Inverted AR Roots	-.00+.71i	-.00-.71i		

Luego de estimar el AR(2) tanto con tendencia como intercepto y que ninguno de significativo llego a esta estimación con coeficiente significativo.



Además las raíces están dentro del círculo unitario.

Date: 10/01/26 Time: 18:07
Sample (adjusted): 1994 2024
Q-statistic probabilities adjusted for 1 ARMA term

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.160 -0.160	0.8709		
		2 -0.078 -0.107	1.0870	0.297	
		3 -0.147 -0.184	1.8733	0.392	
		4 -0.124 -0.207	2.4539	0.484	
		5 -0.092 -0.221	2.7877	0.594	
		6 0.237 0.107	5.0919	0.405	
		7 0.092 0.088	5.4522	0.487	
		8 0.074 0.111	5.6949	0.576	
		9 0.040 0.166	5.7691	0.673	
		10 -0.350 -0.239	11.726	0.229	
		11 -0.087 -0.132	12.109	0.278	
		12 -0.077 -0.234	12.429	0.332	
		13 0.102 -0.139	13.026	0.367	
		14 0.204 0.048	15.527	0.276	
		15 -0.002 -0.122	15.528	0.343	
		16 -0.093 -0.019	16.119	0.374	

Y sus residuos también son ruido blanco.

Como obtengo dos modelos con ruido blanco utilizo el criterio de información de Schwarz para decidir entre ambos, sin embargo primero necesito que ambos tengan la misma cantidad de observaciones para compararlo, por lo tanto debo estimar el modelo ARIMA(0,1,2) con 31 observaciones como tiene el ARIMA(2,1,0).

Dependent Variable: D(LN_PRECIO_PRODUCTOR_USD_GH)
Method: ARMA Conditional Least Squares (Gauss-Newton / Marquardt steps)
Date: 10/01/26 Time: 18:42
Sample: 1994 2024
Included observations: 31
Failure to improve likelihood (non-zero gradients) after 22 iterations
Coefficient covariance computed using outer product of gradients
MA Backcast: 1992 1993

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.043624	0.007929	5.501444	0.0000
MA(2)	-0.966247	0.050524	-19.12448	0.0000
R-squared	0.450141	Mean dependent var		0.057105
Adjusted R-squared	0.431180	S.D. dependent var		0.276678
S.E. of regression	0.208671	Akaike info criterion		-0.233776
Sum squared resid	1.262763	Schwarz criterion		-0.141261
Log likelihood	5.623529	Hannan-Quinn criter.		-0.203618
F-statistic	23.74079	Durbin-Watson stat		2.471016
Prob(F-statistic)	0.000036			
Inverted MA Roots	.98	-.98		

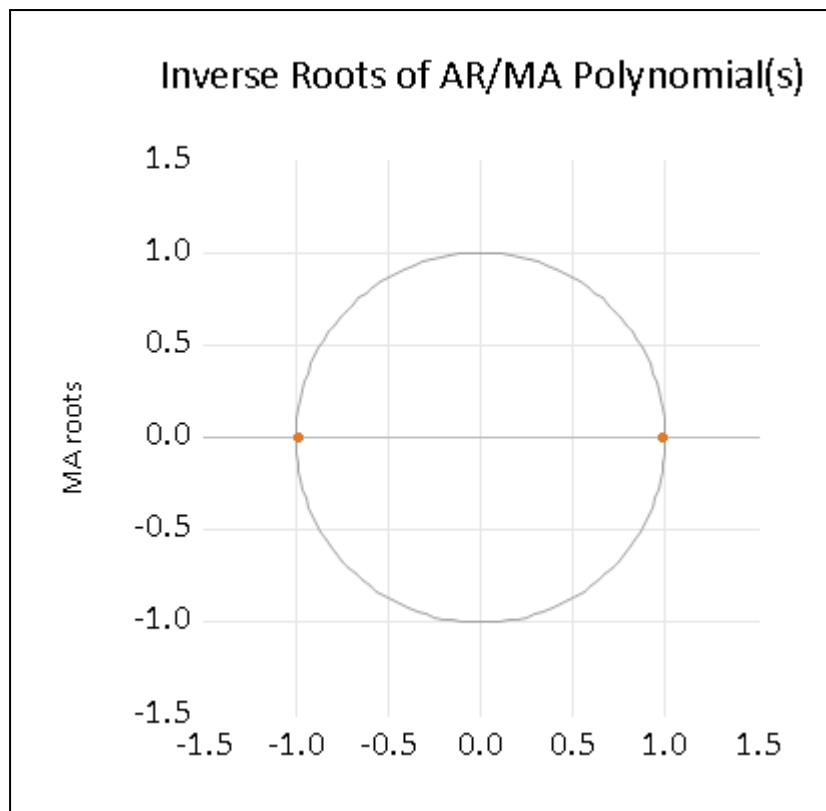
Date: 10/01/26 Time: 18:43

Sample: 1994 2024

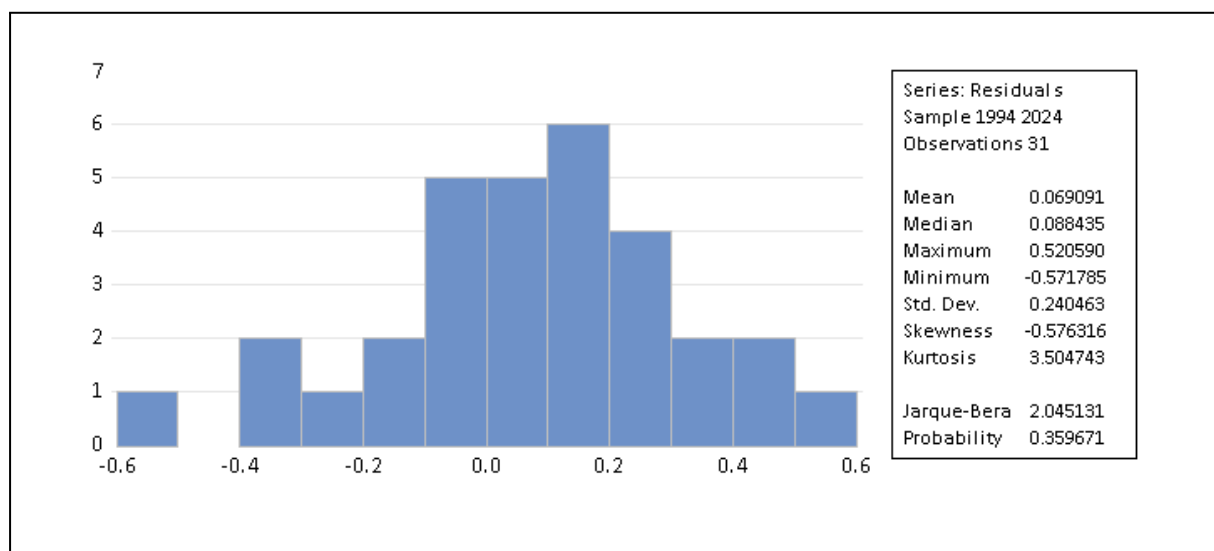
Included observations: 31

Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob
		1 -0.167	-0.167	0.9531	0.329
		2 -0.030	-0.060	0.9850	0.611
		3 -0.048	-0.065	1.0694	0.784
		4 -0.069	-0.094	1.2478	0.870
		5 -0.013	-0.049	1.2540	0.940
		6 -0.093	-0.123	1.6103	0.952
		7 0.018	-0.038	1.6248	0.978
		8 -0.003	-0.034	1.6252	0.990
		9 -0.107	-0.145	2.1561	0.989
		10 0.102	0.032	2.6649	0.988
		11 -0.060	-0.071	2.8501	0.993
		12 0.133	0.091	3.8006	0.987
		13 -0.100	-0.089	4.3675	0.987
		14 -0.036	-0.071	4.4462	0.992
		15 -0.015	-0.068	4.4606	0.996
		16 -0.054	-0.074	4.6585	0.997

La nueva estimación del ARIMA(0,1,2) con 31 observaciones sigue siendo significativa, sigue teniendo ruido blanco e incluso tiene un mejor criterio de Schwarz que el AR(2) sin embargo al utilizar menos observaciones las raíces del modelo se acercaron mucho a la unidad como puede verse en el siguiente gráfico:

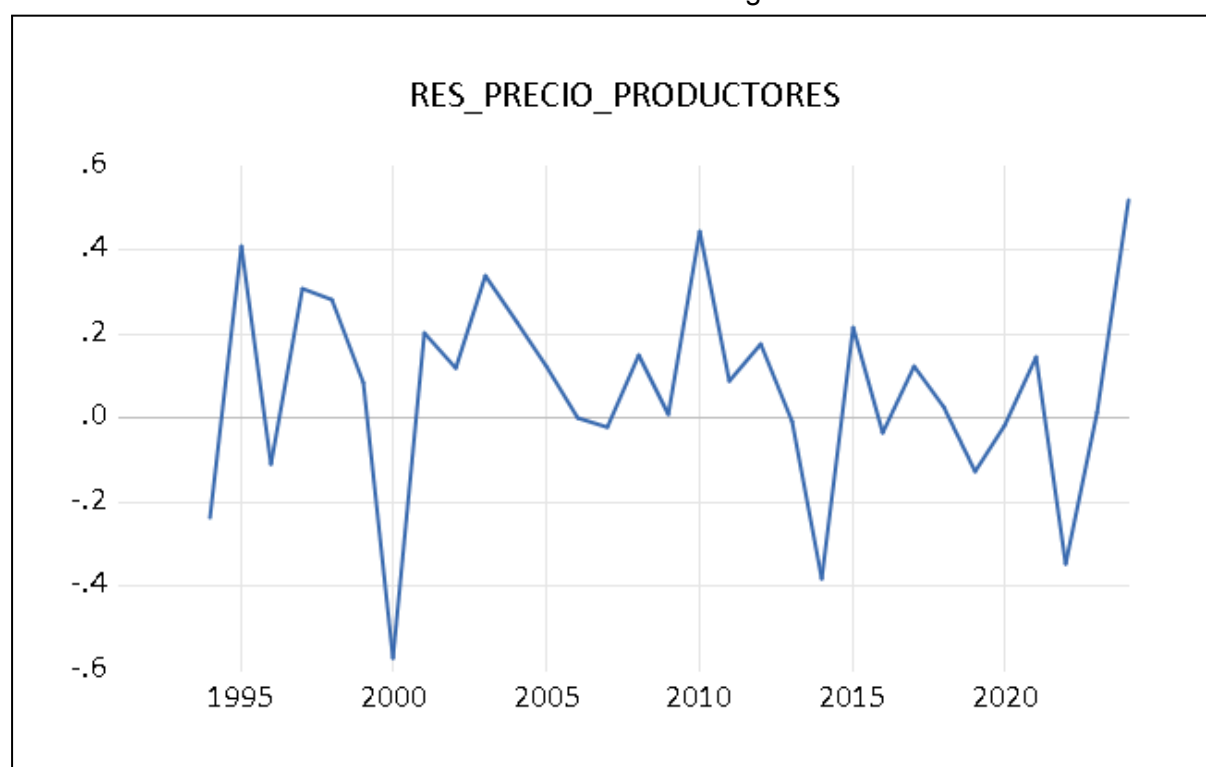


Por lo tanto el modelo seleccionado es el ARIMA(2,1,0).



El test de Jarque-Bera dice que no se puede rechazar la hipótesis nula de que los residuos se comportan normalmente.

Utilizo los residuos del ARIMA(2,1,0) y creo una nueva serie a partir de estos, que como son ruido blanco va a ser estacionario en torno a una media igual a 0.



Date: 10/01/26 Time: 18:50

Sample (adjusted): 1994 2024

Included observations: 31 after adjustments

Autocorrelation		Partial Correlation		AC	PAC	Q-Stat	Prob
		1	-0.160	-0.160	0.8709	0.351	
		2	-0.078	-0.107	1.0870	0.581	
		3	-0.147	-0.184	1.8733	0.599	
		4	-0.124	-0.207	2.4539	0.653	
		5	-0.092	-0.221	2.7877	0.733	
		6	0.237	0.107	5.0919	0.532	
		7	0.092	0.088	5.4522	0.605	
		8	0.074	0.111	5.6949	0.681	
		9	0.040	0.166	5.7691	0.763	
		10	-0.350	-0.239	11.726	0.304	
		11	-0.087	-0.132	12.109	0.355	
		12	-0.077	-0.234	12.429	0.412	
		13	0.102	-0.139	13.026	0.446	
		14	0.204	0.048	15.527	0.343	
		15	-0.002	-0.122	15.528	0.414	
		16	-0.093	-0.019	16.119	0.445	